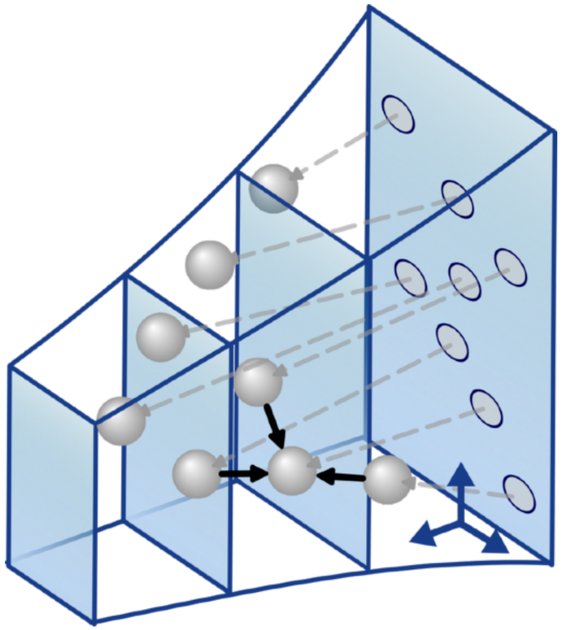
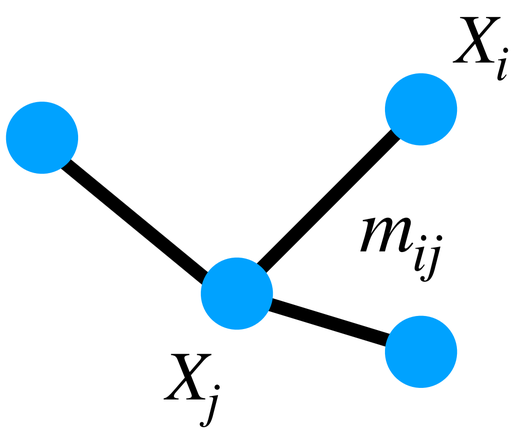
AdS GNN Step 2: Details on bulk operations





We now operate on the points in the bulk using standard **graph neural network** ideas:

1. Each **node** at position X_i has scalar degrees of freedom h_i and is connected to k nearest neighbours by **edges**.

2. Each **edge** carries messages m_{ij} derived from the data on the nodes that they connect.

3. Layers work with **message-passing** as follows:

$$m_{ij}^{(l)} = \psi_e^{(l)}(h_i^{(l)}, h_j^{(l)}, D(X_i, X_j))$$
$$h_i^{(l+1)} = \psi_h^{(l)}(h_i^{(l)}, m_{ij}^{(l)} \,|\, i \in \mathcal{N}(j))$$

where **D is AdS-invariant distance**. At the end, take an average (or other permutation-invariant operation) over all nodes.